Summative Assessment

Resource title:	**Collector Game for a Younger Whānau Member**
Course:	8DGT · Year 8 · Term 4 2026
Progress outcomes:	CT PO3 and DDDO PO2
Resource reference:	8DGT CT3/DDDO2 A v2

Student/Ākonga instructions

Introduction/Kupu Arataki

You will turn a starter collector game into a game for a younger child (about 5 to 10) in your whānau. This child is your end-user (the person who will use what you make). Then you will show your game on a Google Site.

This must be your own work. You must not use generative AI (AI that makes code, words or pictures). Read the Plagiarism poster before you start.

You have 8 periods over 4 weeks. Your kaiako will check your work on 5, 16 and 24 November. Fill in your template as you go. The rubric at the end shows each grade. The expected standard has a star.

Task/Hei Mahi

Keep everyone safe. Keep your end-user's name, photo and school, and your surname, off your site.

No younger child? Ask your kaiako. A classmate can act as your end-user. Home play-tests need an adult there.

Part 1: Plan (Week 1)

1. Talk to your end-user. Find out their age, the games they like, and what they find hard.
2. Write at least three needs (things your game must do for them), such as big sprites.
3. Open the starter game at `arcade.makecode.com/S60890-47146-48977-41276`. Play it, then click **Edit Code** to make your own copy.
4. Plan your changes. Add hazards, lives, a way to win and lose, a sound, a use for **A**, and your own feature.
5. Break your changes into at least three parts (decompose them), such as hazards, lives and your feature.
6. Write an algorithm for your own feature (numbered steps in order, that a computer could follow).
7. Predict what will happen on screen when your steps run.
8. Think of a second way to make your feature work. Write which way you will use and why.

Template: your end-user, needs, changes, algorithm, prediction and second way.

Part 2: Build (Weeks 2–3)

1. Name your project after your game. Change the sprites and theme to suit your end-user.
2. Add your changes one part at a time. Follow your plan.

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3. Add a use for the **A** button (an input). Add outputs such as lives, sounds and a game over screen.
4. Add loops (blocks that repeat code) for two or more jobs, such as making hazards appear.
5. Inside a loop, add a rule that changes the game. For example: if score is more than 10, make the hazards faster.
6. Add each problem to your debugging table: why it happened and how you fixed it.
7. At each checkpoint, click **Share** and post the link privately on the Classroom task.
8. Pick one loop you added. Explain what it repeats, and what would happen without it.

Template: debugging table and one loop explained (no screenshots of code needed).

Part 3: Test (Week 3)

1. Test every control: arrows, **A**, and **B** if used. Write what you expected and what happened.
2. Test the limits: screen edge, last life and win point. If you win at 10, test 9, 10 and 11.
3. Fix anything that fails. Then test again to check nothing else broke.
4. Watch your end-user play. Write what they said and did, and if the controls were easy.
5. Make at least one change because of their feedback.

Template: testing table and end-user feedback table.

Part 4: Showcase (Week 4)

1. Make a Google Site named after your game. Give it three pages: "About", "How to play" and "Behind the game".
2. On "About", say what your game is and who it is for. Put your game on this page too.
3. To add your game, click **Share** in MakeCode and copy the embed code. In Google Sites, click **Insert › Embed › Embed code** and paste it.
4. On "How to play", say what each button does. Explain how your code turns each button press into what the player sees.
5. On "Behind the game", add two or more screenshots and what you changed from the starter.
6. Choose words and pictures that suit your end-user and their whānau.
7. Publish your site. Open it on another device. Check that the game plays and every link works.
8. Explain why you chose each file type and Sites feature. Compare two options for one, such as PNG or JPG.
9. Explain one good and one bad way your game could affect your end-user. Name one choice you made because of it.
10. Explain how children's games have changed since your parents were young, and how that changes their effect on families.
11. Describe a time two needs clashed, such as more levels and short play time. Say what you chose.

Template: steps 8 to 11 (no screenshots of your site needed).

Submitting your work

Check your template is complete, with your game and site links at the top. Hand in on Google Classroom by **Thursday 26 November**. Your work is judged as it is on that date.

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Assessment rubric/Mahere Aromatawai: Collector Game for a Younger Whānau Member

The expected standard for this task is **4P** (marked ★). Each strand gets the grade its evidence best fits. Each grade includes everything in the grades to its left.

Strand	3P	3A	4B	4P ★	4A	Grade
Decomposition and algorithms CT PO3 Evidence: Part 1 and the game	Names the changes they will make. Writes some steps for their own feature, but some are missing or out of order.	Breaks their changes into at least three parts. Writes steps for their own feature in the right order.	Writes a complete algorithm for their own feature, with no missing steps. The code does what the steps say.	Predicts what the algorithm will make happen on screen. Describes a second way to make the feature work.	Explains which way suits the end-user better and why. The code uses that way.	
Inputs, outputs and sequence CT PO3, DDDO PO2 Evidence: Game (share link), How to play page and Part 3	Changes the starter's sprites, and the game still runs in an order that makes sense.	Adds one new output, such as a sound or lives, that works.	Adds a use for the A button. Adds two new outputs (such as lives, sound, game over) that work as planned.	The game can now be won or lost, and ends. How to play explains how each button press becomes what is shown, with the player in control.	Also judges how easy the controls are for the end-user, and explains a change made to make them easier.	
Iteration CT PO3 Evidence: Game (share link) compared with the starter, and Part 2	Adds a loop with help, and it runs.	Adds a loop that does a job in the game, such as making hazards appear.	Adds a loop that works, and says in words what it repeats.	Adds loops for two different jobs. Predicts what would happen without one of them.	Also puts a rule with a comparison inside a loop to change the game. For example, enemies speed up when score is more than 10.	
Testing and debugging CT PO3, DDDO PO2 Evidence: Part 2, Part 3	Plays the game to find problems and fixes at least one, with help.	Records at least two errors in the debugging table and how each was fixed.	Tests every control and records what was expected and what happened.	Also tests the limits (screen edge, win or lose point, last life) and fixes what fails. Improves the game after end-user testing.	Explains why each error happened and how the fix works. Retests to check nothing else broke.	
Making and sharing the showcase DDDO PO2 Evidence: Published site and Part 4	Makes a Google Site with the game's name and a little information about it.	The site is published and shows the game playing (embedded). It says who the game is for.	Has at least three pages with screenshots, and the game plays on another device.	Words and images suit the end-user and their whānau. Explains why each file type and Sites feature was used.	Compares two options for one choice (such as PNG or JPG). Explains why the chosen one suits the purpose better.	
End-users and impact CT PO3, DDDO PO2 Evidence: Part 1, Part 4	Says who the game is for.	Describes the end-user (age, what they like) and one thing they need from the game.	Lists at least three needs of the end-user. The changes meet at least two of them.	Explains one good and one bad effect on the end-user, and a choice made because of it. Says how children's games have changed over time.	Also weighs a trade-off between two things the end-user needs. Explains what was chosen and why.	

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The descriptors are indicative. Judge equivalent evidence in each ākonga's own context.

Overall grade

Grade	What this looks like overall	Overall
3P	A changed starter game that runs, made with help. Planning, testing and the site are started but not finished.	
3A	The starter has new sprites, a new output and an added loop. A simple plan, some debugging, and a published site showing the game.	
4B	New features built from a complete algorithm, with A and new outputs working. Controls are tested. The three-page site works on another device.	
4P ★	A game changed for a real end-user, with a predicted algorithm and a second way. Added loops do different jobs, limits are tested, and feedback improves the game. The site explains choices and impact.	
4A	As for 4P, plus a comparison rule inside a loop and errors explained. Choices are weighed against the end-user's needs, including how easy the controls are.	

Judge only what ākonga added or changed from the starter. The overall grade is a best fit across the six strands, not an average.

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